

Electric Vehicle Charging Management Application

Software Requirements Specification Document

1. Introduction

1.1 Purpose

The **Electric Vehicle Charging Management Application (EV CMA)** is software developed for a single charging-network provider.

Its purpose is to enable the provider to offer and manage electric-vehicle charging services through web and smartphone applications, while maintaining administrative control through a command-line interface.

The system allows electric-vehicle users to:

- Create accounts
- Locate and reserve charging stations
- Start and stop charging sessions
- Pay securely through integrated payment services

For the provider, the application supports:

- Data management
- Pricing configuration
- Monitoring of station status
- Reporting

Overall Goal:

Provide a unified and efficient software solution that improves the charging experience for users and streamlines operations for the provider.

Stakeholders: User (EV Driver), Admin, External Services (Payment Service, Energy Market, Authentication Service, Map Service, Chargers)

1.2 Interfaces

1.2.1 Interfaces to External Systems

The **Electric Vehicle Charging Management Application (EV CMA)** communicates with external systems for payment, real-time pricing, map integration, and data sharing with mobility platforms.

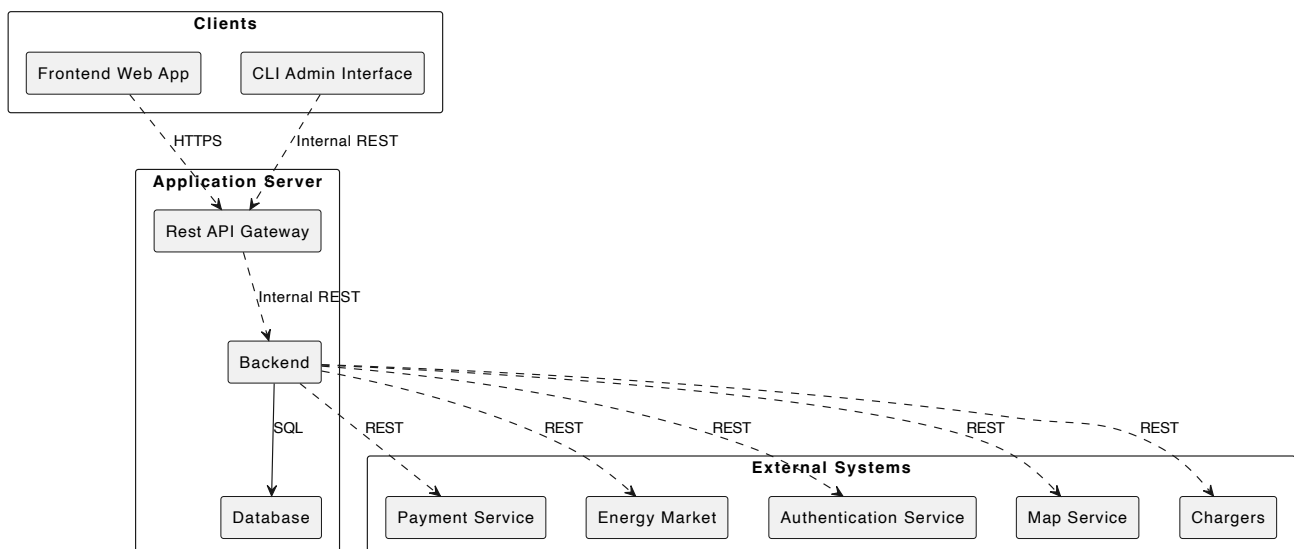
Identification of External Systems

External System	Description
Payment Service Provider (PSP)	Processes user payments and refunds via secure REST API.
Energy Market Service	Provides live wholesale pricing data for dynamic tariff calculation.
Charging Infrastructure / Chargers	Network of connected EV chargers exchanging real-time status, start/stop commands, and metering data with the system.
Authentication Service	Manages user identity, login tokens, and authorization policies for both clients and backend components, using OAuth 2.0 / OpenID Connect standards.
Mapping and Routing APIs	Used to display and navigate to charging stations (e.g., Google Maps, OpenStreetMap).
User Devices	Web browsers and mobile apps connecting through HTTPS to the provider's REST API.

Applicable Standards and Protocols

- **RESTful HTTP / HTTPS**
- **OAuth 2.0 / OpenID Connect**
- **JSON** (data format)
- **TLS 1.2 or higher** (security)
- **PCI-DSS** — for payment security

UML Component Diagram



1.2.2 User Interfaces

View and Reserve

EV CMA - Map View

[Search chargers...]

Filters: Connector | Power | Price | Available

• Charger-01

• Charger-02

• Charger-03

MAP

[Center]

[List]

[Charger Status]

[Profile]

[Help]

Charger Details

Station NTUA-01
Address: Zografou Campus
Distance: 1.2 km
Status: AVAILABLE

Connectors: Type2, CCS
Max Power: 22 kW
Tariff now: 0.42 €/kWh

[Reserve Charger]

[Navigate to Charger]

[Back]

Reserve Charger

Select duration (max 30')
() 10 min () 20 min (X) 30 min

Estimated pre-authorization: 15.00 €

[Confirm Reservation]

[Cancel]

Reservation Active

Charger: Charger-01
Time left: 00:18:45

COUNTDOWN / TIMER BAR

[I arrived / Check-in] [Navigate to charger]

[Cancel Reservation]

[Back to Map]

Charging and Details

Charging Session

Energy delivered: 6.4 kWh
Time elapsed: 00:18:32
Estimated cost: 2.69 €
Remaining preauth: 12.31 €

LIVE PROGRESS / CHART

[Navigate to charger]

[Report Issue]

[Back to Map]

Session Summary

Final energy: 12.8 kWh
Total time: 00:41:10
Final cost: 5.22 €
Payment: Captured

[Download Receipt]

[Share]

[Go to Profile Stats]

[Navigate to charger]

[Report Issue]

[Back to Map]

Profile & Statistics

Petros P.

[Edit]

Email: petros@mail.com

[Edit]

Payment: Visa **** 1234

[Edit]

Sessions	Energy	Cost
14	168 kWh	72.4 €

TREND CHARTS

[Detailed History]

[Back to Map]

Detailed History

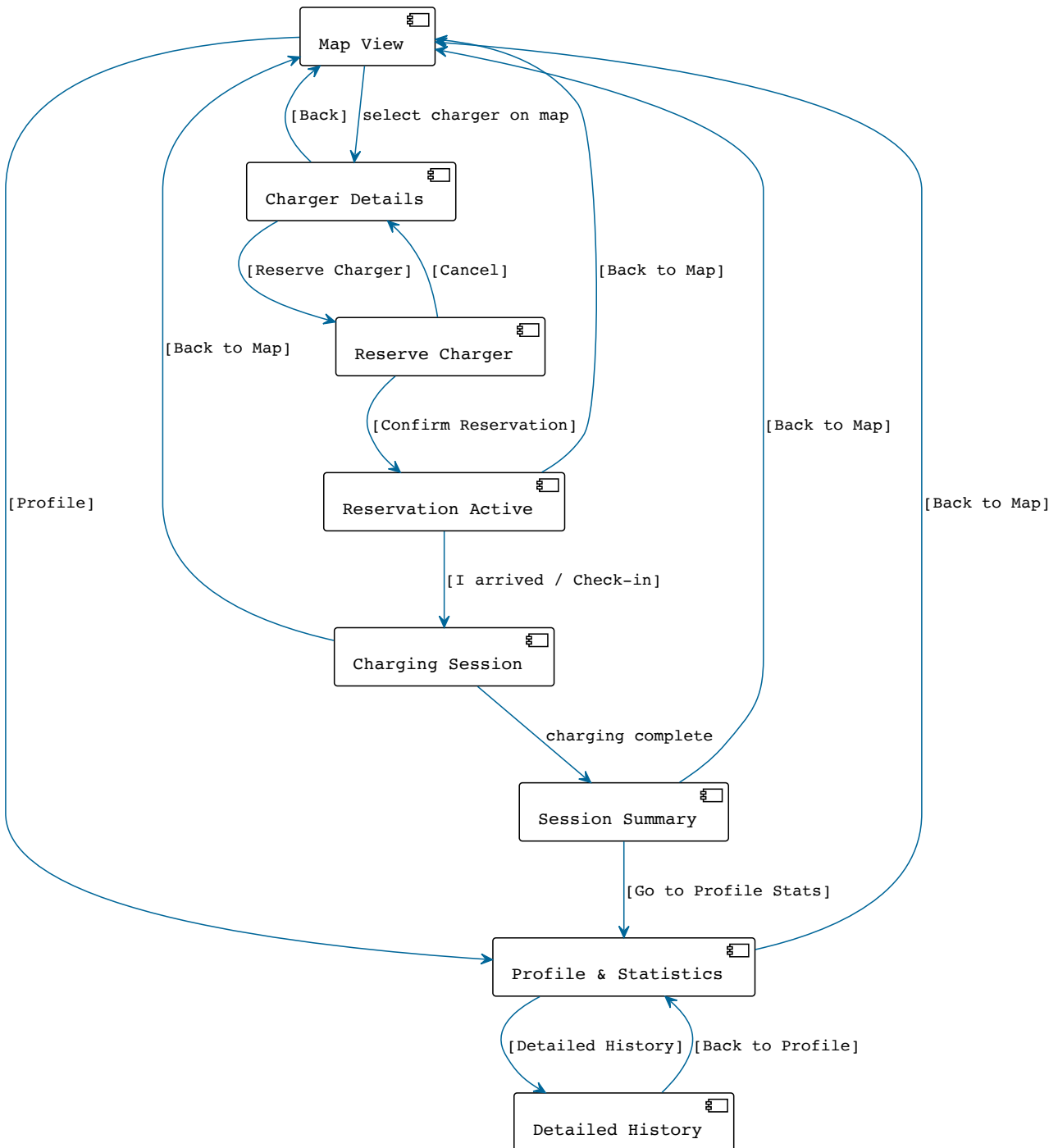
Date range: [Last month v]
Charger: [All chargers v] [Apply] [Reset]

Date	Station	Charger	Energy	Cost
date1	station1	Charger-01	12.3 kWh	5.18 €
date2	station1	Charger-02	8.7 kWh	3.65 €
date3	station1	Charger-01	15.0 kWh	6.28 €
...	...	...	...	...

[Back to Profile]

[Back to Map]

EV CMA - Screen Navigation Flow



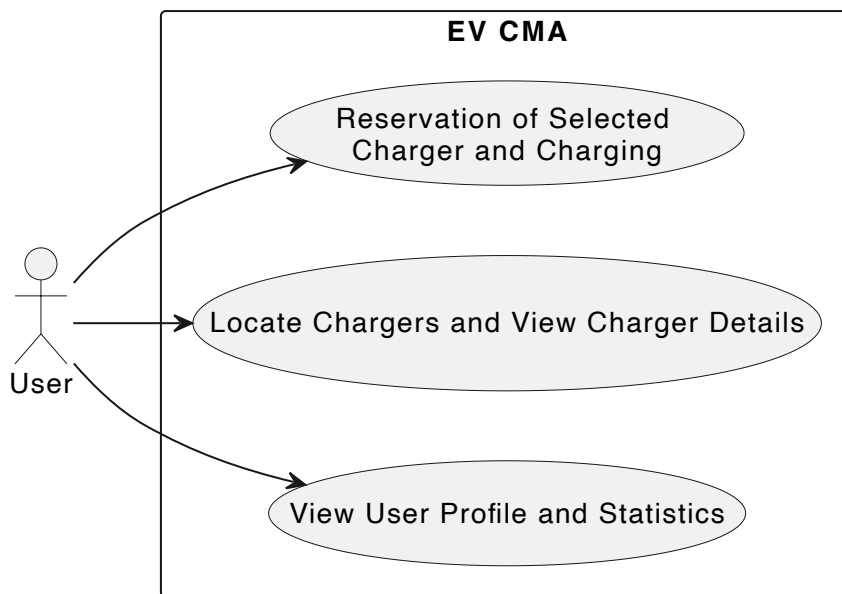
2. References

[
- Links to information sources.
- Glossary of terms
]

3. Software requirements

3.1 Use cases

3.1.0 Use Case Diagram



3.1.1 Use case 1: Reservation of Selected Charger and Charging

This use case covers the full workflow of reserving an available charger, initiating charging after successful payment pre-authorization, monitoring the session, and finalizing payment and session data upon completion.

3.1.1.1 Roles involved

- User

3.1.1.2 Preconditions

- The application must be running.
 - The user is authenticated and has an active session in the system.
 - A specific charger has been selected by the user and is currently available for reservation.
 - The user has valid payment information stored or available for use.

3.1.1.3 Execution environment

[
- Web frontend, smartphone ui, api execution context - For user interfaces: add wireframes as plantuml/salt, mermaid.js/"user journey", or D2
]

3.1.1.4 Input data

[
- user input, or input for other sources, if applicable
]

3.1.1.5 Expected behaviour

Step 1: Reservation Initiation The user selects the option "Reserve Charger" from within the application. The system retrieves the available time slots (e.g., a maximum of 30 minutes) and prompts the user to select the desired reservation duration. **Step 2: Duration Selection** The user chooses the desired reservation duration. The system verifies that the selected duration is within the allowed limits (e.g., cannot exceed 30 minutes). If the selection is invalid, the user is asked to select again.

Step 3: Availability Re-Check Before finalizing the reservation, the system performs a real-time availability check for the selected charger.

- If the charger is still available, the reservation is confirmed.
- If the charger is no longer available (e.g., taken by another user), the system informs the user and prompts them to select a new charger.

Step 4: Reservation Waiting Period Once the reservation is confirmed, the user is given a time window to reach the charger. The system displays the remaining time until reservation expiration. If the user does not arrive before the countdown ends, the reservation is automatically canceled.

Step 5: User Arrival & Payment Pre-Authorization Upon arrival at the charger, the user checks in by confirming presence via the application. At this point, the system initiates payment pre-authorization using the user's selected payment method.

- If the pre-authorization succeeds, the user may proceed.
- If it fails, the system allows the user to retry using another payment option or cancels the session.

Step 6: Start Charging Session If the payment is successfully pre-authorized, the user selects the option "Start Charging". The system activates the charger and the charging session begins. Progress information (energy delivered, charging time, cost estimation) is displayed to the user in real time.

Step 7: Charging Session Monitoring While the charging session is active: - The user may choose to manually stop charging. - The user's battery may become fully charged. - The prepaid amount (if applicable) may be exhausted.

During charging, the system continuously updates the user with real-time progress (charging percentage, energy delivered, time, cost). These updates are also sent to the backend.

Step 8: Charging Session Termination Charging stops when:

- The user manually ends the charging session, or
- The system automatically ends the session (battery full, prepaid amount used up, or any other system condition).

Once charging ends, the system:

- Calculates the final cost based on the actual energy consumed.
- Captures the final payment amount.

Step 9: Post-Charging Processing After charging is completed and the final payment is processed, the system:

- Calculates the total charging time.
- Calculates the total cost.
- Issues the final receipt to the user.
- Stores the charging session in the user's charging history.

Alternate flows

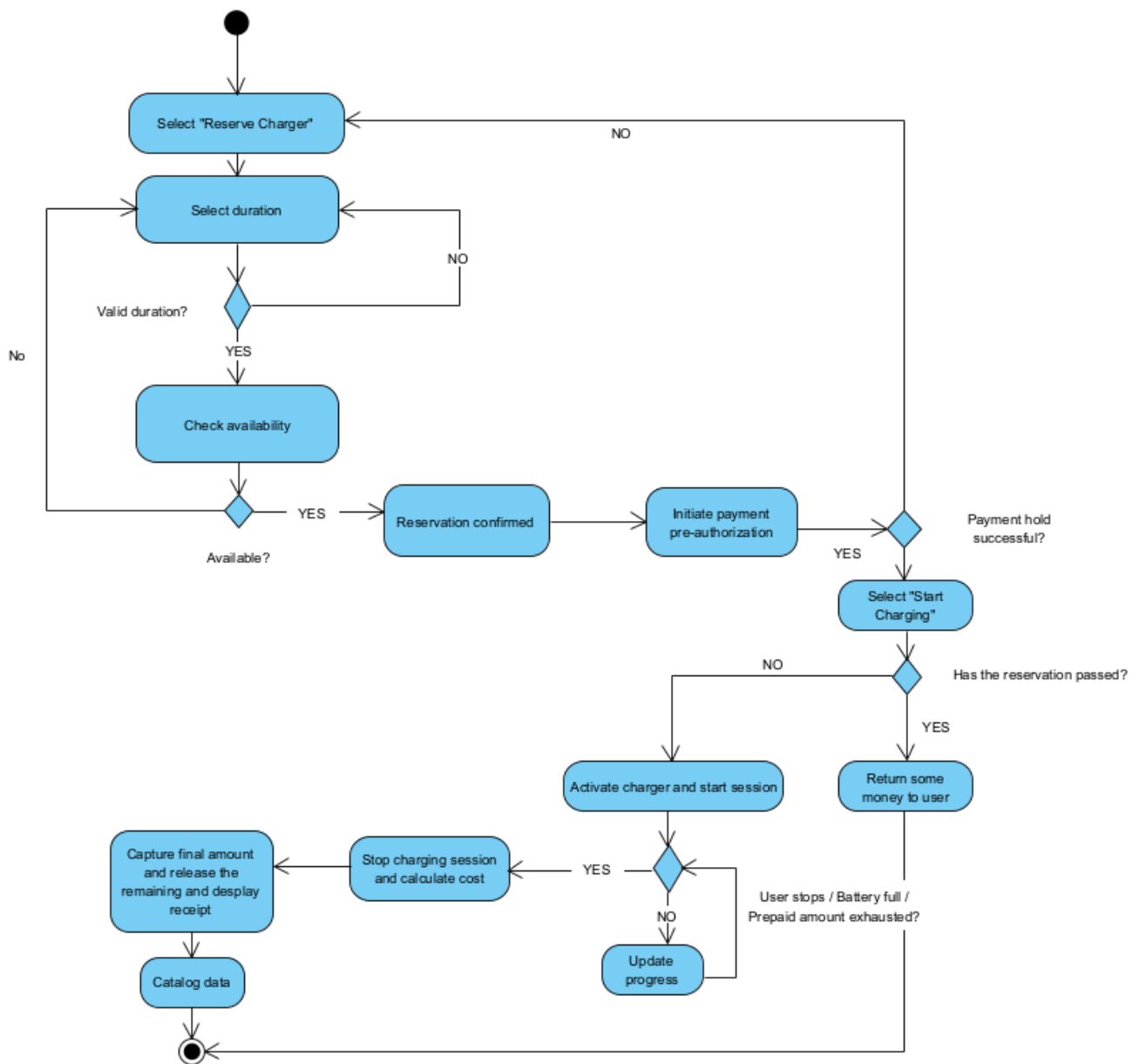
- A1 – User selects invalid reservation duration
System prompts the user to re-enter a valid duration.
- A2 – Charger becomes unavailable before confirmation
System notifies the user and suggests selecting another charger.
- A3 – Payment pre-authorization fails
User may retry with a different payment method. If all attempts fail, the session is cancelled.
- A4 – User does not arrive before the reservation expires *Reservation is cancelled automatically. System notifies the user.*

3.1.1.6 Output data and postconditions

- The charging session has ended (either by user action or automatically) and the charger is released and marked as available for other users.
- The final energy consumption, total charging time and total cost have been calculated and stored in the system.
- The user's payment has been processed successfully (or the session has been cancelled if payment failed).
- A digital receipt has been issued and is available to the user.
- The completed charging session has been added to the user's charging history and can be used for future statistics and reporting.

3.1.1.7 Notes

First Case Activity Diagram



3.1.2 Use case 2: Charger Discovery and Selection

This use case covers the workflow of discovering available chargers on the map, applying filters to refine search results, selecting a charger, and viewing detailed information about the selected charger.

3.1.2.1 Roles involved

- User

3.1.2.2 Preconditions

- The application must be running.
 - The user is authenticated and has an active session in the system.
 - The system has access to the database of chargers and their status.
 - Location services or map access permissions (if required) are enabled.

3.1.2.3 Execution environment

[
- Web frontend, smartphone UI, API execution context - For user interfaces: add wireframes as plantuml/salt, mermaid.js/"user journey", or D2
]

3.1.2.4 Input data

[
- User input (e.g., filter criteria, charger selection)
]

3.1.2.5 Expected behaviour

Main flow

Step 1: Map Initialization The user opens the application (Web or Mobile). The system retrieves the user's location (if applicable) and loads the main map view, displaying chargers within the region.

Step 2: Apply Filters The user applies filters (e.g., charger type, plug type, power rating, availability). The system requests updated charger data from the backend based on the selected filters.

Step 3: Display Filtered Chargers The system displays updated markers on the map, showing only the chargers that meet the filtering criteria. If no chargers match the filters, the system informs the user.

Step 4: Select a Charger The user selects a charger from the map. The system identifies the charger ID and fetches detailed information from the backend.

Step 5: View Charger Details The system displays full charger information, including:

- Charger status (available, in use, offline)
- Technical specifications (power rating, connector type, charging speed)
- Estimated cost for charging
- Distance from the user
- Address and navigation option

The user may proceed to the next actions (e.g., reserving the charger — handled in Use Case 1).

Alternate flows

- A1 — Backend fails to return charger data
The system displays an error message and prompts the user to retry.
- A2 — No chargers match the applied filters
The system displays a message indicating "No chargers found," and offers the option to reset or modify filters.
- A3 — Selected charger becomes unavailable before details are loaded
The system notifies the user that the charger is no longer available and updates the map.

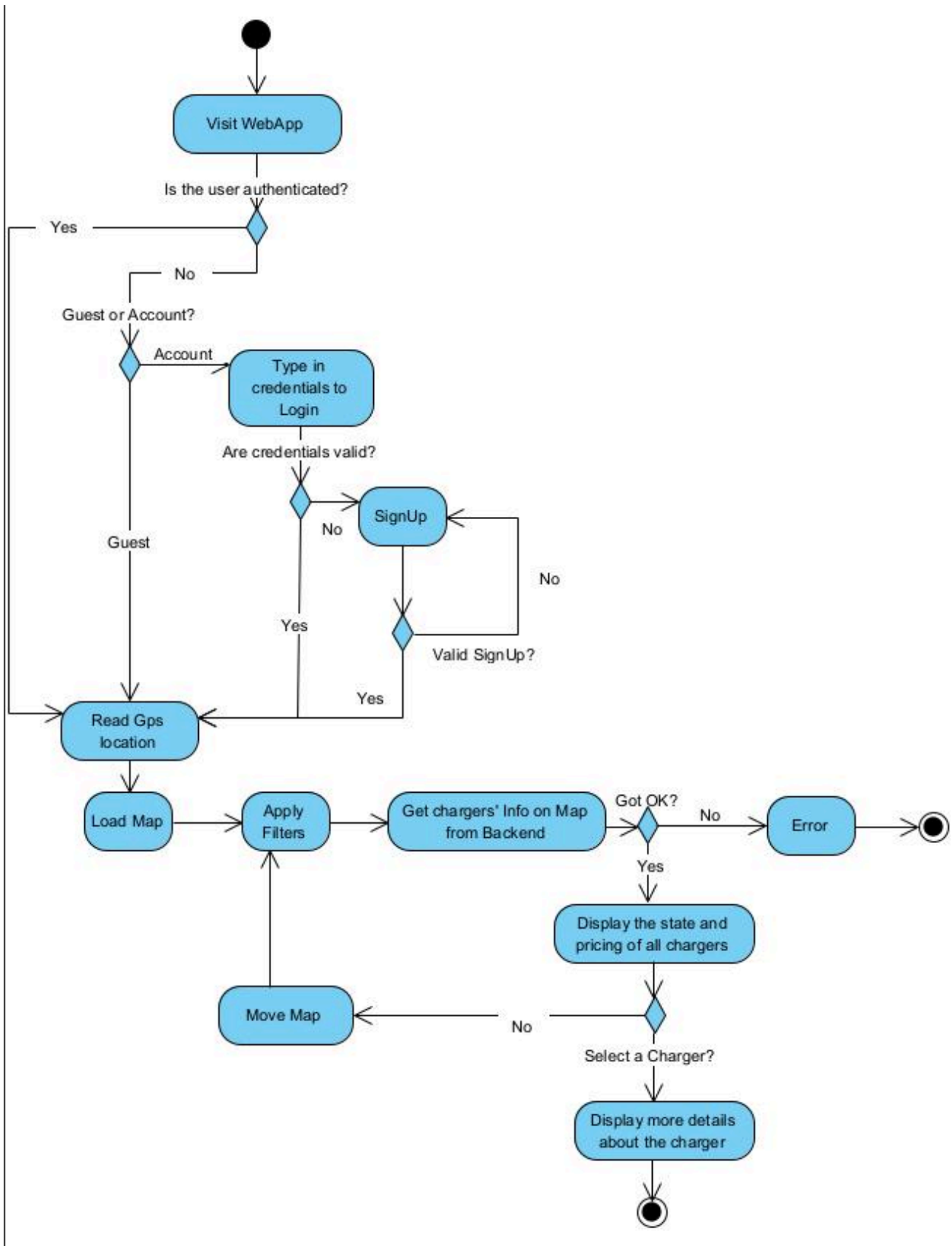
- A4 — Location services unavailable (for mobile use)
The map loads without user location and displays chargers only based on a broader view or user-adjusted map area.

3.1.2.6 Output data and postconditions

- The system has successfully displayed chargers on the map according to the user's selected filters.
- The user has viewed detailed information about the selected charger, including availability, technical specifications, and estimated cost.
- Any updates to charger status or availability have been reflected in the system's map view.
- The user is able to proceed seamlessly to further actions (e.g., reservation in Use Case 1) based on the displayed charger information.
- The system has logged any errors encountered (e.g., failed data retrieval) for backend monitoring and diagnostics.
- The user may proceed to the next steps, such as making a reservation or starting a charging session.
- A digital receipt has been issued and is available to the user.
- The completed charging session has been added to the user's charging history and can be used for future statistics and reporting.

3.1.2.7 Notes

Second Case Activity Diagram



3.1.3 Use case 3: User Profile and Charging Statistics

This use case describes how an authenticated user accesses their profile page and views both their basic account information and their charging statistics. The system retrieves the user's details and historical charging data, validates access, and displays summarized and detailed statistics such as past sessions, total energy consumption, cost summaries, and long-term usage trends.

3.1.3.1 Roles involved

- User

3.1.3.2 Preconditions

- The application must be running.
 - The user is authenticated and has an active session in the system.
 - The system has access to the user's stored profile information.
 - The statistics module has access to historical charging data for the user.

3.1.3.3 Execution environment

[
- Web frontend, smartphone UI, API execution context - For user interfaces: add wireframes as plantuml/salt, mermaid.js/"user journey", or D2
]

3.1.3.4 Input data

[
- User input (e.g., selecting profile or statistics section)
]

3.1.3.5 Expected behaviour

Main flow

Step 1: Access Profile Section The user selects the "Profile" or "Account" option from the application menu (web or CLI). The system loads the profile settings page and prepares to fetch user-related information.

Step 2: Retrieve Basic Profile Information The system requests the user's profile data (name, email, preferences, payment methods, etc.) from the backend. If retrieval is successful, the system displays the basic profile information.

Step 3: Request User Statistics In parallel with displaying the basic profile data, the system initiates a request to fetch the user's charging statistics from the backend. These may include:

- Number of charging sessions
- Total energy consumed
- Total cost paid
- Average session duration
- Monthly or long-term summaries

Step 4: Display Statistics If the statistics retrieval is successful, the system displays:

- Historical charts
- Consumption summaries
- Cost breakdown

- Comparative or long-term usage patterns
This enables the user to review their energy usage, expenses, and charging behavior.

Step 5: User Navigation The user may choose to:

- Return to the main page
- Review more detailed data
- Modify account settings
- Or exit the profile section

The use case ends when the user leaves the profile/statistics page.

Alternate flows

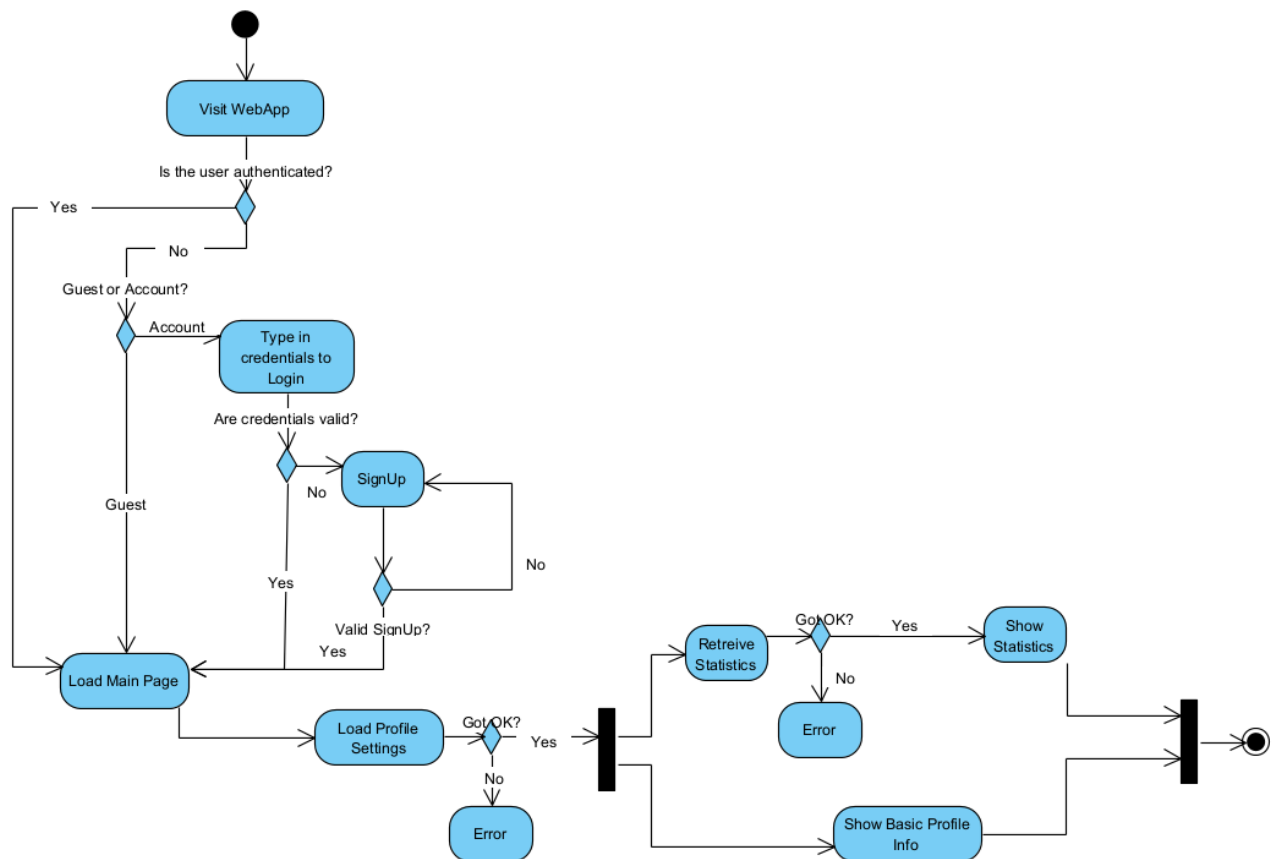
- A1 — Failed Retrieval of Profile Information
The system displays an error message and prompts the user to retry. Basic profile information is not displayed until retrieval succeeds.
- A2 — Failed Retrieval of User Statistics
*If the system cannot retrieve statistics:
Only basic profile information is shown.
The statistics section displays an error or "No data available."
The user may retry the operation.*
- A3 — No Historical Data Available
*If the user has not completed any charging sessions:
The statistics section displays "No charging history found."
The user is still able to view basic profile information.*

3.1.3.6 Output data and postconditions

- The user's basic profile information has been successfully retrieved and displayed, provided backend access was available.
- Charging statistics have been retrieved and displayed, or the system has indicated that no data is available or an error occurred.
- The system has updated the profile and statistics interface with the most recent data stored in the backend.
- Any failed retrieval attempts (profile or statistics) have been logged for backend monitoring and diagnostics.
- The user has exited the profile/statistics section and returned to another part of the application or closed the session.

3.1.3.7 Notes

Third Case Activity Diagram



3.2 Functional Requirements

[
- UML functional requirements diagram
]

3.2.1 Functional UI Requirements Specification

FR-1: User Login

The system shall allow users to log in using valid credentials and authenticate their identity before accessing any personalized functionality.

FR-2: User Profile Selection

The system shall allow authenticated users to access and manage their personal profile settings, including preferences, payment methods, and vehicle details.

FR-3: Fetch User Statistics

The system shall retrieve and display key usage statistics for the user, such as past charging sessions, total energy consumed, and cost summaries.

FR-4: Reserve Charger

The system shall allow users to reserve an available charger for a selected time period and ensure the charger remains locked for their use.

FR-5: Payment Handling

The system shall process payment operations, including pre-authorization, final cost calculation, and charging the user based on actual energy consumption.

FR-6: Start Charging

The system shall enable users to initiate a charging session once they arrive at the selected charger and the payment validation has been completed.

FR-7: Timer Expiration → Notify User

The system shall notify users when their reservation time is about to expire or has expired, prompting them to either extend or release the charger.

FR-8: User Registration

The system shall allow a new user to sign up (create account with credentials, basic info, car info, etc.).

FR-9: View Chargers on Map/List

The system shall show available chargers and their prices on a map (or list) around the user's GPS position or a selected area.

FR-10: Search / Filter Chargers

The system shall allow the user to apply filters (e.g. connector type, power, price range, distance, availability) and update the map/list accordingly.

FR-11: View Charger Details

The system shall display detailed info for a selected charger (location, connectors, max power, current status, pricing rules, opening hours, etc.).

FR-12: Navigation to Charger

The system shall be able to hand off the charger location to an external navigation app (e.g. Google Maps) or built-in navigation.

FR-13: Manage Payment Methods & Preferences

The system shall allow the user to add/remove/update payment methods and set defaults or charging preferences (e.g. max prepaid amount).

FR-14: Charging Session Monitoring

During an active session, the system shall periodically update and display progress (energy delivered, elapsed time, estimated cost, remaining prepaid amount).

FR-15: User-initiated Stop Charging

The system shall allow the user to stop an ongoing charging session at any time and calculate the cost up to that point.

FR-16: Session History & Bills

The system shall provide detailed session history and bills for the last 6 months and aggregated statistics for longer periods.

FR-17: Data Cataloging of Sessions

The system shall store all charging sessions (start/end time, charger, energy, cost, user) for later reporting and statistics.

3.2.2 Functional Admin-CLI Requirements Specification

FR-18: Admin Login via CLI *The system shall provide a secure CLI interface where an admin can authenticate with admin credentials.*

FR-19: Add New Charger (CLI) *Through the CLI, the admin shall be able to add a new charger (location, capabilities, tariffs, initial status) with validation and database update.*

FR-20: Modify Charger Configuration/Status (CLI) *Through the CLI, the admin shall be able to modify charger info (tariffs, availability, technical parameters) and update the database.*

FR-21: View Charger Status (CLI) *The CLI shall allow querying the current status of a charger (online/offline, in use, fault codes) and display the information.*

FR-22: Retrieve Network Statistics / Reports (CLI) *The CLI shall allow the admin to retrieve and display aggregated statistics (number of sessions, total energy, revenue, usage per charger, etc.).*

FR-23: Publish Charger Status to Third-party Services *The system shall expose the status of chargers to authorized third-party applications (e.g. public EV maps) via an appropriate API.*

3.3 Performance Requirements

PR-1: Response Time (UI) *For 95% of requests, the system shall respond to user actions, such as loading the map, searching chargers, or reserving a slot, within 2 seconds under normal load.*

PR-2: Map Loading Time *The initial map view with charger data around the user shall be fully interactive within 3 seconds after login or page load.*

PR-3: Charger Data Refresh Rate *The system shall refresh charger status (availability, in-use, offline) every 30 seconds to ensure up-to-date information.*

PR-4: Concurrency Handling *The system shall support at least 500 concurrent user sessions without degradation beyond the specified response-time thresholds.*

PR-5: Reservation Conflict Resolution *The system shall guarantee atomicity of reservation operations. Two users cannot reserve the same charger simultaneously.*

PR-6: Payment Processing Throughput *Payment processing (authorization → confirmation) shall complete within 5 seconds in 95% of cases, given normal network conditions.*

PR-7: Scalability *The system shall be designed to scale horizontally to handle up to 5× baseline user load without major architectural changes.*

PR-8: Data Consistency and Latency *All user or charger status updates shall propagate to all relevant views and APIs within 2 seconds of completion.*

PR-9: API Response Time (for Third-party Integrations) *The public API for charger status shall deliver responses within 1 second for up to 50 requests per second.*

PR-10: Resource Utilization *During normal operation, average server CPU utilization shall stay below 70% and memory utilization below 75%, ensuring capacity for traffic peaks.*

3.4 Data Requirements

3.4.1 Data access requirements

The system accesses and manages data through a combination of internal components, external services, and physical charger devices. This section describes the sources of data, technologies used, communication protocols, and authentication mechanisms. [

--Data sources and sinks--

-Relational database

Stores all persistent domain data (users, chargers, reservations, charging sessions, tariffs, payments, analytics). Accessible only through the backend service.

-Backend service

Acts as the single access layer for all data. Provides REST endpoints for frontend, mobile clients, and CLI.

-EV chargers

Provide operational status and telemetry (availability, energy delivered, power level). Receive commands such as "start session" and "stop session." Connected through provider-specific protocols (e.g., HTTP(S), MQTT, or vendor API).

-Payment service provider

Handles pre-authorisation, final charging, and refunds. Exposes a secure web API accessed exclusively by the backend.

-External information providers

Provide dynamic elements such as wholesale electricity prices. Optionally interact with navigation services (Google Maps, etc.).

-CLI (Provider operations)

Accesses data via REST API for most operations and via dedicated admin endpoints for provider-only tasks.

--Technologies and communication protocols--

-REST API layer

Communication over HTTPS JSON request/response format HTTP methods used appropriately (GET/POST/PUT/DELETE)

-Database access layer

Backend communicates with SQL database via an ORM or driver No direct database access from client applications

-Device integration layer

Backend abstracts low-level charger communication Uses HTTP(S), MQTT, WebSockets, or vendor-specific protocols (team's design choice)

-External service integration

Payment provider accessed with API keys Dynamic pricing provider accessed with secure HTTPS requests

--Authentication and authorisation--

-User authentication

Username/password login Session represented by token (JWT or session ID) Required for all user-specific API operations

-Provider/admin access (CLI)

Separate authentication mechanism for staff Role-based access control enforced at API level

-Service-to-service authentication

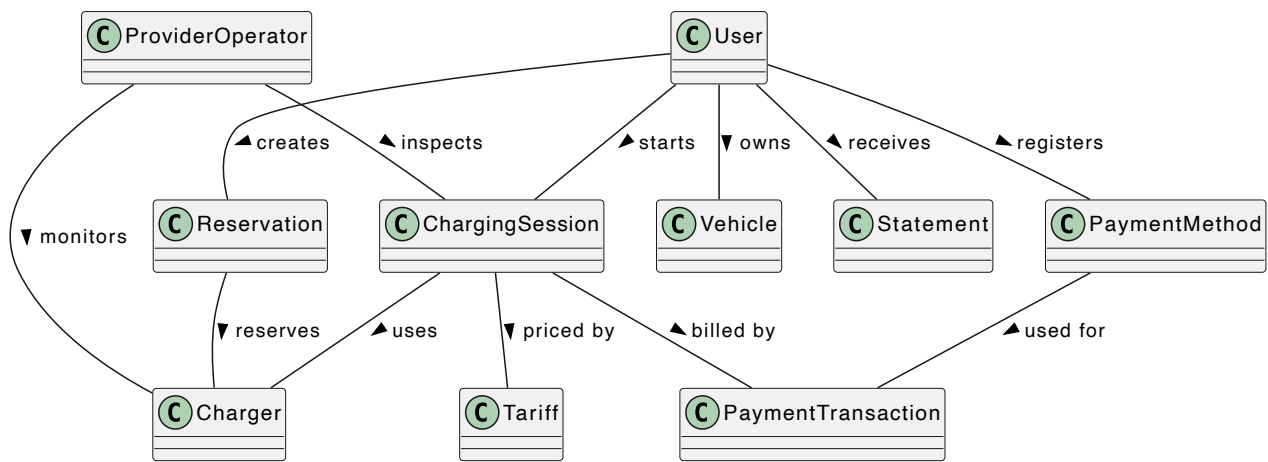
API keys or client secrets for payment and pricing services Secrets stored securely (environment variables or vault)

--Security requirements--

All communication involving user or payment data must use TLS/HTTPS Passwords stored only as salted hashes Payment card data never stored—only tokens from the payment provider Logs must not contain sensitive personal or financial information]

3.4.2 Semantic data model

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3.5 Other requirements

3.5.1 Availability

AR-1: *The system shall maintain a minimum 99.5 % uptime per month, excluding planned maintenance not exceeding 30 minutes per month.*

AR-2: *Planned maintenance windows shall be scheduled outside peak usage hours and announced at least 24 hours in advance.*

AR-3: *The system shall support automatic failover between redundant servers to minimize user disruption.*

AR-4: *Backup procedures shall guarantee full restoration of operational data (users, chargers, sessions, payments) within 30 minutes in the event of system failure.*

AR-5: *All service components (web server, API, payment gateway) shall be continuously monitored for health and availability.*

AR-6: *In case of communication failure with a charger, the system shall retry for up to 60 seconds and mark the charger status as temporarily unavailable if retries fail.*

3.5.2 Security

SR-1: *All client–server communication shall use TLS 1.3 or higher to protect data in transit.*

SR-2: *Sensitive data (passwords, payment tokens, personal information) shall be stored with AES-256 encryption or equivalent.*

SR-3: *User passwords shall be stored in a hashed and salted format following bcrypt or a comparable algorithm.*

SR-4: *The system shall implement role-based access control (RBAC) separating user, operator, and admin privileges.*

SR-5: *The admin CLI shall require multi-factor authentication and log all executed administrative commands.*

SR-6: Sessions shall automatically expire after 15 minutes of inactivity.

SR-7: Payment processing shall comply with PCI-DSS standards and use secure, tokenized transactions. No credit-card details shall be stored locally.

SR-8: The system shall sanitize all user input to protect against SQL injection, cross-site scripting (XSS), and related vulnerabilities.

SR-9: The system shall provide mechanisms for account recovery (email verification and password reset) without exposing sensitive information.

SR-10: All data handling shall comply with GDPR or relevant local privacy regulations.